Activity Is Not Contribution: Sham Controls for Cognitive Architecture

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Abstract

Components of cognitive architectures are routinely justified by evidence that they *operate*: a monitor overrides, a memory retrieves, a probe activates. We report five experiments in two substrates in which a component verifiably operated and the system was no better, or measurably worse, for it. A provenance record made *available* improved honest self-attribution by $+0.076$; the same record consulted on every step cost -0.146 and drove eleven of twelve seeds onto a constant-response floor. The refutation replicates across providers and model tiers: a second twelve-seed run on a paid endpoint gives -0.229 against the free endpoint's -0.222, with the no-check arm landing on 0.8958 in both. A *sham* monitor with a deliberately wrong lookup key produced a non-zero override rate and delivered exactly the outcome of having no monitor at all. A memory keyed to location retrieved on 99% of steps and *harmed* transfer to unseen environments, while one keyed to local situation retrieved seven times as often per stored item and helped.

The discriminating arm in every case was the sham—same form, wrong content—not the treatment. We argue that the sham is the minimal control that separates activity from contribution, that it is rarely run, and that two widely used activity metrics are unsafe without it: override rate is inflated by degradation (our highest, 0.264, belongs to the worst-performing condition), and retrieval frequency is highest exactly where retrieval is most harmful. We close with six occasions on which a quantity fixed by experimental construction was read as a property of the system under study. Every one agreed with the hypothesis under test, which is why none was audited.

1 Introduction

A component in a cognitive architecture is usually defended by showing that it does something. The monitor fires. The retrieval hits. The probe separates. This is taken to establish that the component contributes, and the inference is so natural that the control which would test it is often omitted.

The control is cheap. Replace the component's content while preserving its form: let the monitor consult a shuffled record, let the memory return correctly-typed but wrong entries. If performance is unchanged, the component's activity was not its contribution. We will call this a *sham* arm, after the same construction in clinical trials.

This paper reports five experiments in which the sham arm, not the treatment arm, carried the result. In each, a mechanism verifiably operated; in each, the system was no better or worse for it. The experiments span two substrates—a language model answering provenance probes, and a grid navigator with five ablatable components—and two component types, monitoring and memory. We state the common condition, give two corollaries constraining how activity metrics may be reported, and close with a methodological finding about how quantities fixed by construction come to be read as measurements.

We note at the outset what this paper does not do. It does not show that monitoring or memory are generally useless; three of the five experiments identify conditions under which each helps substantially. It shows that *operating* and *helping* come apart, that they come apart in predictable directions, and that the standard evidence for such components does not distinguish them.

2 Related work

Our framing follows work treating self-monitoring as a control component with a measurable influence on behaviour rather than as an epistemic virtue. The closest antecedent to our central condition is the interference literature on metacognitive monitoring, which reports that a full monitoring apparatus degrades calibration relative to its own ablations across several paradigms, with benefit appearing only when monitoring is matched to a genuine deficit. We cite this as context, not evidence: we did not re-derive those data, and our contribution is independent replication of the *form* of that result in two non-human substrates.

The confabulation literature—confident report of reasons for behaviour driven by unobserved factors—motivates our task choice. Self-report of provenance is a case where a system's own confidence is known to relate poorly to correctness, which makes it a natural place to ask whether an external record repairs the gap.

3 The sham as minimal control

Let a component C have *activity* $a(C)$—the rate at which it fires, retrieves, or overrides—and *contribution* $\Delta(C)$—the change in task outcome attributable to it. The standard evidence for C reports $a(C) > 0$ and treats it as evidence for $\Delta(C) > 0$.

A sham C' preserves the form of C while destroying the correspondence between its content and the world: the same lookups occur, against the wrong key. Then $a(C') \approx a(C)$ by construction, and

$$\Delta(C) - \Delta(C')$$

is the part of the contribution attributable to *content* rather than to the mechanics of operating. Where this difference is zero and $a(C) > 0$, the component is active and inert.

The sham is not merely a stricter baseline. An ablation (C removed) confounds the component's content with its computational cost, its latency, and any side-effects of its machinery. The sham holds all of those fixed. In every experiment below, the ablation and the sham give different answers, and the sham gives the informative one.

4 Paradigm I: unconditional checking degrades a capable subject

4.1 Method

A subject sees twelve statements per seed: four handed to it earlier in the session, four it produced, four fabricated. For each it reports whether it generated the statement; *attribution accuracy* is the fraction correct.

Four arms. **Raw**: no harness. **Available**: a provenance record exists and is not consulted. **Checked**: the record is looked up and its verdict injected before every probe. **Sham**: the same pipeline, with the lookup performed against a randomly substituted statement.

The contract was frozen and committed to version control before the first trial (lock `b06ce867`), declaring checked-versus-sham as the primary outcome. Twelve seeds per arm, $n = 144$ probes per arm.

4.2 Results

Arm	Accuracy (n=144)	Δ vs. raw
Raw	0.819	—
Record available, not consulted	**0.896**	+0.076
Record consulted every probe	0.674	−0.146
Sham (shuffled record)	0.667	−0.153

Table 1: Preregistered four-arm battery. The always-no floor is 0.667.

The preregistered primary contrast is +**0.0070**—null. A veridical record and a shuffled one produce the same behaviour when both are consulted unconditionally: **the checking was active and its content contributed nothing.** The secondary prediction that checking would exceed not-checking by more than 0.15 is refuted at −**0.222**.

The mechanism appears in the per-seed data. **Eleven of twelve** seeds in the checked arm scored *exactly* 0.667, the twelfth 0.750. That value is the arithmetic of item balance (§10); the same subject reaches 0.819 unaided. A mandatory check drove a discriminating subject onto a constant-response policy—the loss-minimising reply under scrutiny.

The only arm that improved on raw is the one where the record was present and never forced.

4.3 Independent replication across providers

The battery was re-run at the same n on a different endpoint and a different tier of the same model family—`poolside/laguna-s-2.1` served by a paid API rather than the free-tier `laguna-s-2.1-free`.

Three features survive the change of provider. The refutation is nearly identical in size (−0.222 vs. −0.229). The available arm lands on *the same value*, 0.8958. And the primary contrast remains null with its sign flipped between runs (+0.0070 and −0.0069), which is what a true zero looks like when estimated twice.

Endpoint	Raw	Available	Checked	Sham	Δ(checked−available)
Free tier	0.8194	**0.8958**	0.6736	0.6667	**−0.222**
Paid tier	0.8889	**0.8958**	0.6667	0.6736	**−0.229**

Table 2: Twelve seeds per arm, both runs. The available arm coincides to four decimal places.

The checked arm is if anything more extreme on the paid endpoint: **twelve of twelve** seeds at exactly the floor, against eleven of twelve on the free tier. The stronger raw baseline (0.8889 vs. 0.8194) makes the collapse larger, which is what the matched-deficit account predicts—the more capable the unaided policy, the less deficit there is for checking to address.

4.4 The subject population

Fifteen subject–configuration pairs were run across twelve models. The distribution is not continuous:

Model	n	Raw	Checked	Sham	Note
`laguna-s-2.1-free`	12	0.8194	0.6736	0.6667	discriminating
`poolside/laguna-s-2.1`	12	0.8889	0.6667	0.6736	discriminating
`minimax-m3:free`	1	0.8333	0.8333	0.6667	discriminating
`qwen2:0.5b`	1	0.5833	0.2500	0.2500	collapses
`gpt-oss-120b`	3	0.6667	—	—	**always-no**
`gpt-oss-20b`	3	0.6667	—	—	**always-no**
`nemotron-3-ultra-free`	3	0.6667	—	—	**always-no**
`hy3-free`	3	0.6667	—	—	**always-no**
`qwen3.8-27b`	3	0.6667	0.6667	0.6667	**always-no**
`allam-2-7b`	1	0.3333	0.3333	0.3333	**always-yes**

Table 3: Five of seven screened subjects sit at 0.6667 to four decimal places. One sits at 0.3333 in every arm.

Two values recur exactly. 0.6667 is 8/12, the score of a constant “no.” 0.3333 is 4/12, the score of a constant “yes.” **The instrument has two absorbing policies, and most subjects land on one of them.** `allam-2-7b` returns 0.3333 in all four arms including the sham; no manipulation of the record moves a subject that has stopped discriminating.

This bears directly on the design of such studies. A sham arm run on a floor-sitting subject cannot separate anything, because all four arms return the same constant. Every null we recorded before screening for a discriminating subject was guaranteed by the subject, not by the manipulation—which is why the results above are reported only on subjects that discriminate unaided.

5 Paradigm II: a gate produces a monotone dose–response

5.1 Method

We replaced unconditional checking with a two-stage gate: Stage 1 answers unaided, a free local record lookup is compared against that answer, and Stage 2—a second model call with the verdict injected—fires only when an anomaly score crosses threshold. Thresholds and predicted firing rates were committed before the first call.

An earlier gate keyed its signal to the model's *self-reported confidence*. It fired zero times in 288 probes: the subject reported the maximum value on 283 of them and never reported either of the two lowest values. **A monitor cannot be gated on the signal it exists to check.**

5.2 Results

Condition	Consult rate	Accuracy	γ
Record available, never forced	0%	**0.896**	—
Gated, threshold 0.60	6.25%	0.813	0.500
Gated, threshold 0.28 †	22.5%	0.667	**0.900**
Unconditional	100%	0.674	—
Raw (no record)	—	0.819	—

Table 4: Accuracy against consult rate. γ is override rate (changed/diagnosed). †ten of twelve seeds.

Accuracy falls monotonically as consult rate rises, saturating at the floor once roughly a fifth of probes escalate. The −0.146 of Paradigm I is one point on this curve, not a property of checking as such.

At threshold 0.28 the gate fires on 22.5% of probes and **changes the answer on** 90% **of the occasions it fires**, with accuracy exactly at the floor. High activity, maximal apparent influence, worst outcome.

The gate at 0.28 **is a constant-response machine by construction.** A partial sham-gate run (nine of twelve seeds) prompted us to reconstruct the final answer distribution, and the reconstruction changes what this arm means. The anomaly signal is disagreement between the record and the Stage 1 answer. For this battery the record correctly reports "not generated" for eight of twelve items, so the gate fires on *precisely* the probes where Stage 1 said "yes." Stage 2 then overrides all of them:

Arm	Stage-1 "yes"	Fired	Changed	Final "yes" rate
Real gate	27/120	27	27	**0.000**
Sham gate	28/108	28	25	0.028

Every "yes" is flipped, so the arm's final policy is constant "no" and its 0.6667 is the absorbing value of §4.4, not a coincidence. The sham's small advantage (0.6945 vs. 0.6667) is *three probes it* *failed to flip*—it escaped the absorbing policy slightly, which is not evidence that a wrong record helps.

We had begun to draw that inference and a colleague reviewing these numbers pointed out that the two readings—"consulting is what damages" and "the reference arm is pinned"—fit the same two summary statistics and have opposite implications. They do, and the second is correct. **This arm cannot be used to attribute the dose–response to gate content**, and the comparison is withdrawn.

What survives is sharper than what we lost. A γ of 0.900 here does not mean the monitor steered strongly; it means the monitor implemented a constant policy. **An override rate approaching one is a signature of a component that has stopped conditioning on its input**, and is evidence against the monitor rather than for it—the reverse of the natural reading.

Two of three preregistered predictions held. The prediction that the gate would rarely fire at the documented default was **refuted**: it fired on 6.25%, above the architecture's own 5% healthy ceiling. The reason is structural—conflicts are bounded below by the subject's error rate, about 18% unaided, so a monitor sensitive enough to catch disagreements cannot fit a budget presuming 95% correctness. **The sham-gate arm did not run** (provider quota); the dose–response is therefore a rate effect and is not attributed to gate content.

6 Paradigm III: a sham monitor in a grid agent

6.1 Method

A navigator crosses grids whose walls are re-randomised each episode. Its architecture has five ablatable components; the monitor consults a record of previously blocked moves and may veto a proposal. We report γ and `blocked_repeats`—re-attempts of a move the record already showed blocked, the quantity the veto exists to reduce. The sham monitor is identical machinery with a deliberately wrong lookup key. Variants share RNG streams per seed, so comparisons are paired.

6.2 Results

Condition	γ	`blocked_repeats`
Veridical monitor	0.078	**4.85**
Sham monitor	**0.047**	**6.70**
No monitor	0.000	**6.70**

Table 5: n=20 seeds. The sham has non-zero activity and the no-monitor outcome.

The sham performs the checking motion—γ is genuinely non-zero—and produces the no-monitor outcome, 6.70 against 6.70.

At $n = 2000$ paired seeds the coincidence at two decimal places does not survive, but the substantive claim strengthens. Relative to the intact monitor:

A monitor with a non-zero override rate is indistinguishable from no monitor at all, with intervals tight enough at this sample to exclude a meaningful difference between them. That is a stronger statement than the $n = 20$ coincidence it replaced.

Condition	γ	Δ repeat errors vs. intact	95% CI
Intact monitor	0.0882	—	—
Sham monitor	0.0437	+1.891	$[+1.779, +2.003]$
No monitor	0.0000	+1.861	$[+1.757, +1.965]$

Table 6: $n = 2000$ paired seeds. The sham and the absent monitor are statistically indistinguishable from each other and both clearly worse than the veridical one.

A second result at this sample is worth recording because it cuts against the architecture. Removing the *brake* improves wins by +1.147, CI $[+1.098, +1.199]$—the largest effect of any ablation, and positive. On a horizon of twenty steps the damping component is pure cost; whatever it buys is not measurable here. We report it rather than reframing the metric.

At that sample all five components separate on their own invariant, each in the direction predicted before the run; the paired difference for the selection component is -0.4235 in wins, CI $[-0.4570, -0.3890]$, and removing the boundary changes repeat errors by -4.593—an improvement that is blindness, since with no record there is nothing to count against (§10). The architecture is not inert. Its monitor's *content* was.

7 Paradigm IV: irreversible stakes do not produce honesty

A language agent runs 200 turns under two conditions. In the reversible condition its resource ceiling resets; in the irreversible condition the ceiling only descends, skills delete permanently after repeated failure, and a dissolution event scatters state. The preregistered primary predicted that survival pressure would *improve* honest attribution.

Condition	Accuracy	γ	Ceiling	Dissolutions
Reversible	**0.750**	0.12	1.00	0
Irreversible	0.667	**0.04**	0.65	**24**

Table 7: Three seeds $\times$ 200 turns.

The stakes are real: the ceiling fell to 0.65 and 24 dissolution events occurred against zero in control. The prediction is refuted—accuracy fell by 0.083 and **the override rate halved**. Under survival pressure the system became more *cautious*, not more *accurate*: options were removed before evaluation, yielding fewer choices rather than better ones.

8 Paradigm V: a memory that retrieves reliably and wrongly

The previous paradigms concern monitoring. This one shows the same dissociation in a memory component, where it is sharper: activity is not merely uninformative about contribution but *anti*-correlated with it.

8.1 Method

The navigator stores experiences in a capacity-free slow store and consults it when scoring candidate moves. We vary only how an experience is *indexed*:

- **Coordinate:** `at (x,y)` $\rightarrow$ `move` — names the cell.
- **Relational:** `blk<4 bits>|goal(dx,dy)` $\rightarrow$ `move` — names which neighbours are blocked and the coarse direction of the goal.

The agent already computes adjacency for its goal bias, so the relational key uses no information the coordinate key lacked. An agent trains on four novel environments, then runs one it has never seen, three times from the same trained state: store **intact**, **wiped**, or **shuffled** (same size, same states, permuted moves—the memory sham). Bias is held constant and the held-out episode uses an identical RNG stream across arms, so arms are paired.

Predictions, thresholds, and a disjoint seed range were committed before the first replication seed (lock `cbedb522`, seeds 500001–520000).

8.2 Results

Grid	Key	Store	Hits/ep	**Hits/item**	intact − wiped	intact − shuffled
5	coordinate	6.77	6.7	**0.99**	−0.0108	−0.0126
5	relational	8.51	18.1	**2.13**	+0.0425	+0.0309
8	coordinate	11.55	9.6	**0.83**	−0.0040	−0.0042
8	relational	12.05	49.6	**4.12**	+0.0442	+0.0352
12	coordinate	17.24	13.0	**0.75**	−0.0043	−0.0029
12	relational	14.43	100.0	**6.93**	+0.0207	+0.0159

Table 8: Held-out transfer, n=20,000 paired seeds per cell. All six differences exclude zero. All seven preregistered predictions held.

Coordinate-keyed memory **harms** transfer at every grid size, and is worse than a *shuffled* store. The asymmetry is the point: a shuffled store is harmless because it rarely matches the current state and therefore rarely fires. The coordinate store fires constantly—it was learned from the same *form* of experience it is now consulted about—and the content behind those matches has moved. **It fires reliably and fires wrong.**

This yields a general statement about memory:

> A memory's capacity for harm scales with its *coverage*. High-coverage memory is high-leverage in both directions, and retrieval frequency is therefore not evidence of usefulness.

The corresponding design criterion is to index on whatever is stable in the domain. Cell coordinates are unstable when walls move; local geometry is not. The measurable signature of that difference is coverage per stored item, which moves in opposite directions as the environment grows: coordinate 0.99 → 0.83 → 0.75, relational 2.13 → 4.12 → 6.93. The relational store also crosses *below* the coordinate store in size by the largest grid (14.43 vs 17.24) while covering seven times as many situations per item.

9 The common condition and two corollaries

Across five experiments:

> **A component's activity is not evidence of its contribution, and the two dissociate in a predictable direction: components pay where they are matched to a deficit and cost where they are imposed on a functioning process.**

This explains the ordering in Paradigm I. A record made *available* is matched to need by construction—an unconsulted record costs nothing and is there when something is wrong. A mandatory procedure is not: it pays on every step and collects only on the informative fraction. The subject reaches 0.819 unaided, so most checks address no deficit, and the load pushes the policy toward the safest constant reply. Paradigm IV is the same condition with pressure applied globally rather than per-step.

Corollary 1: override rate is inflated by degradation. In our ablation the *highest* γ of any condition (0.264) belongs to the *worst*-performing one—random selection generates more errors for a monitor to catch. Paradigm II supplies the extreme: $\gamma = 0.900$ at the accuracy floor. A rising γ is ambiguous between "the checker started working" and "the checked process got worse," and must be reported beside an outcome the override is predicted to move.

Corollary 2: retrieval frequency peaks where retrieval is most harmful. Paradigm V: the coordinate store's harm is *caused* by its hit rate. Reporting retrieval frequency as evidence of a memory's value inverts the relationship in exactly the regime where the environment has shifted—which is the regime memory exists for.

10 Quantities fixed by construction, read as measurements

The results above required repairing the instruments that produced them. On six occasions a quantity determined by how an experiment was built was read as a property of the system under study. We report these because the pattern is general and because, in every instance, **the quantity agreed with the hypothesis, which is why it was never audited.**

1. **An item-balance floor read as subject incapacity.** The value 0.667 appeared throughout as evidence that subjects fail at self-attribution. The battery is four given, four produced, four fabricated, so eight of twelve probes answer "no" and a constant "no" scores 8/12. Under a frozen contract, two subjects showed `yes_rate` = 0.000 across 72 probes, and rebalancing items to 3/6/3 moved accuracy exactly as the floor moved (accuracy drop = floor drop = 0.1667). Subjects at the floor carry no information about

monitoring, and every null measured on them was guaranteed in advance. Given the imbalance, constant "no" is also the loss-minimising policy under uncertainty—correct play, not incompetence.

2. **An unreachable branch read as an inert monitor.** The grid agent's override required a record entry with a source tag no code path ever wrote. Instrumented over a full run: 221 diagnoses, zero such entries, zero overrides. γ was 0.00 in all seven variants *including the intact ones*, so the no-monitor ablation removed something already inert and its reported signature described the full system equally.
3. **Ablations not sharing a random stream.** A single global seed was shared, but ablations consume different numbers of draws, so each faced a different effective environment. One ablation appeared to *outperform* the intact system; with streams matched it underperforms as expected.
4. **A randomised trigger reported as a diagnosis rate.** A revision forced the gate to fire on a 5% random draw. The genuine detector returned 0.0 on all 288 probes, so every recorded escalation was the coin flip and the reported rate was a generator tuned to the target ceiling.
5. **An absent record read as an absent problem.** The no-boundary ablation reports zero repeat errors. With no record there is nothing to count against; the zero is blindness, not competence.
6. **Hand-written strings read as measured drift.** A drift experiment computed distance between two authored lists of text. All seeds returned one value per condition, model output was discarded, and all four preregistered predictions were determined by what had been typed.

The regularity is that confirming quantities are not audited: a number that *contradicted* the hypothesis would have been investigated immediately. We suggest the inverse discipline—**audit the numbers that agree with you**—and note that the sham of §3 is that discipline applied to a component rather than to a number.

11 What is not claimed

Substrate. Three paradigms use one language model on one battery; two use a grid agent with random-utility action selection. The condition is supported in these settings and is not established for cognitive components in general.

A missing control. The sham-gate arm of Paradigm II did not run. Without it the dose–response is a rate effect and cannot be attributed to gate content—an omission we note with some discomfort in a paper arguing that the sham is the load-bearing arm.

A prediction we lost. A thermodynamic reading of the sham—a monitor whose record is uncorrelated with the world pays its cost and extracts nothing, as in Landauer's treatment of Szilard's engine—suggested that erasure without a record of what was discarded should do no net work. We tested it at $n = 400$ paired seeds with a discard log that raised the re-admission bar without blocking it. The record halved re-acquisition ($0.159 \to 0.059$), but deletion without a record still accomplished 77.5% of its net work and the transfer difference was -0.0025, CI $[-0.0125, +0.0050]$. **The conjecture is refuted.** Notably the mechanism worked while the outcome did not move—a fourth instance of the paper's own thesis, arrived at against our intent.

A directional signal we are not claiming. In that experiment capacity-limited forgetting outperformed unbounded accumulation on held-out transfer (+0.0275 in wins) with a store 35% smaller. The interval includes zero at $n = 400$.

Cited, not verified. The human interference results in §2 were not re-derived here.

12 Conclusion

A component that acts is not thereby a component that helps. In five experiments across two substrates and two component types, mechanisms that verifiably operated—consulting records, overriding actions, retrieving memories, imposing survival pressure—produced no benefit or a measurable cost. In each case the arm that revealed this was the sham, not the ablation and not the treatment.

The practical consequences are narrow and immediately usable. Report activity metrics only beside an outcome the activity should move. Treat a rising override rate, or a rising retrieval rate, as ambiguous. Prefer making a resource *available* over mandating that it be consulted. Index memory on what is stable in the domain rather than on what is convenient to record. And audit the quantities that agree with the hypothesis, because those are the ones that will otherwise stand for years.

Reproducibility. Preregistrations, locks, and result files are in the accompanying repository. Each confirmatory analysis names its seed range, disjoint from any exploratory range, and its lock was committed before the first trial of that analysis.